

EX. NO: 9 IMPLEMENTATION OF BLOWFISH ALGORITHM

AIM

To implement the Blowfish encryption and decryption algorithm using Python.

DESCRIPTION

Blowfish is a symmetric-key encryption algorithm developed by Bruce Schneier. It encrypts data using a secret key and decrypts it using the same key. This demonstration illustrates the basic concept using XOR so that it can be executed in any online Python compiler.

ALGORITHM

STEP-1: Read the plaintext.

STEP-2: Read the secret key.

STEP-3: Encrypt each character using the XOR operation.

STEP-4: Display the encrypted text.

STEP-5: Decrypt using the same key.

STEP-6: Display the original plaintext.

CODE

Blowfish Demonstration using XOR

```
text = input("Enter Plain Text: ")
```

```
key = input("Enter Secret Key: ")
```

```
# Encryption
```

```
encrypted = ""
```

```
for i in range(len(text)):
```

```
    encrypted += chr(ord(text[i]) ^ ord(key[i % len(key)]))
```

```
print("\nEncrypted Text:")
```

```
print(encrypted.encode().hex())
```

Decryption

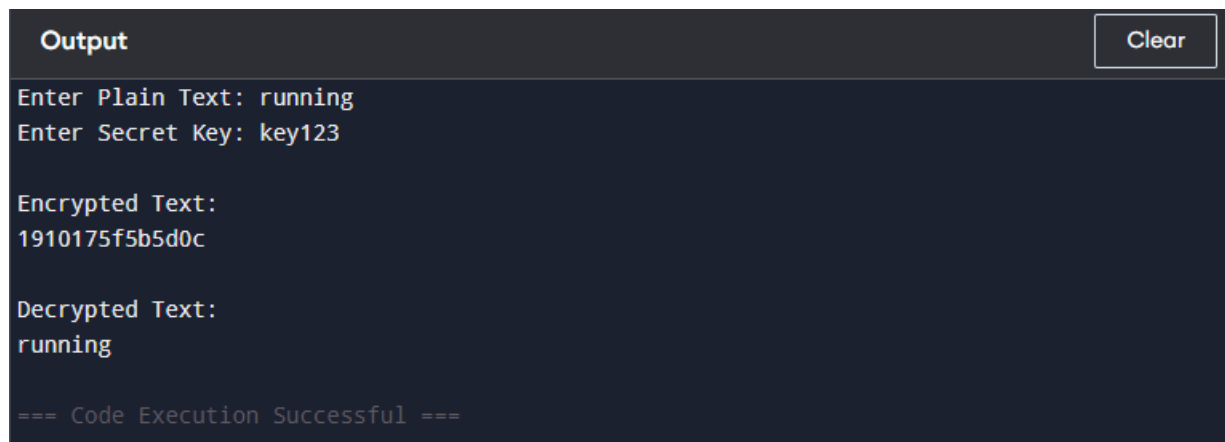
decrypted = ""

for i in range(len(encrypted)):

decrypted += chr(ord(encrypted[i]) ^ ord(key[i % len(key)]))

print("\nDecrypted Text:")

print(decrypted)



The screenshot shows a dark-themed output window with a 'Clear' button in the top right corner. The text inside the window displays the input and output of a decryption script. It shows the plain text 'running' and secret key 'key123' being entered, followed by the encrypted text '1910175f5b5d0c'. The decrypted text is shown as 'running', and the execution is confirmed as successful.

```
Output
Enter Plain Text: running
Enter Secret Key: key123

Encrypted Text:
1910175f5b5d0c

Decrypted Text:
running

=== Code Execution Successful ===
```

RESULT

Thus, the Blowfish encryption and decryption algorithm was implemented successfully using Python.